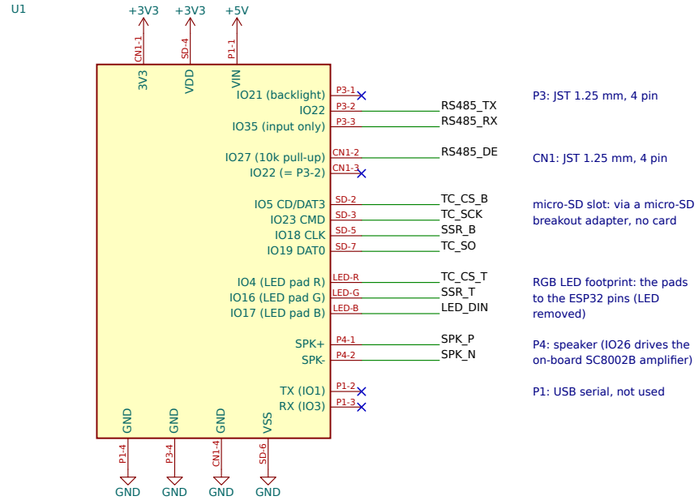
CYD ESP32-2432S028R (RGB LED removed)



P3: JST 1.25 mm, 4 pin

CN1: JST 1.25 mm, 4 pin

micro-SD slot: via a micro-SD breakout adapter, no card

RGB LED footprint: the pads to the ESP32 pins (LED removed)

P4: speaker (IO26 drives the on-board SC8002B amplifier)

P1: USB serial, not used

Notes

TempCtl Duo, env cyd-2xlocal-2x485; pins as in src/board_cyd.h (firmware 1.0.39). Modules connected with cables: a connection schematic of the whole device, not a PCB layout.
Net labels with the same name are connected.

Power: USB on the CYD. 3V3 for the modules from CN1-1 or the SD VDD pin (the board's regulator), 5 V for the SSR inputs from P1-1 (VIN). One ground for the CYD and every module.

The SD-slot pins: plug a micro-SD breakout (sniffer) adapter into the slot; no SD card. IO5 has the board's SD pull-up (converter deselected at boot); IO18 has none (SSR off with the 10k pull-down).

RGB LED removed: wire to the pads that go to the ESP32 pins, not the ones to the 1k resistors to 3V3 (check with an ohmmeter, or remove those resistors too). IO4 floats at boot: R1 keeps the top converter deselected.

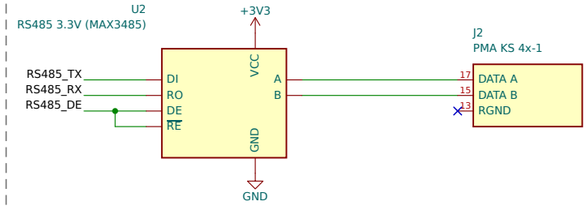
No E-stop input, no SSR supply relay and no chip temperature on the CYD (no free pins). IO35 is input only.

A software controller is never the only protection: fit an independent over-temperature limiter and a contactor.



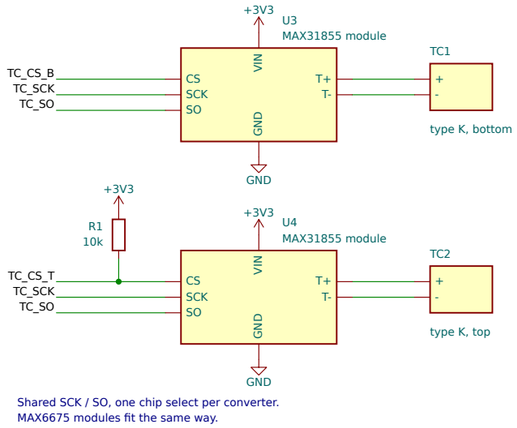
one ground for the CYD and every module

RS485 (Modbus RTU, beta)



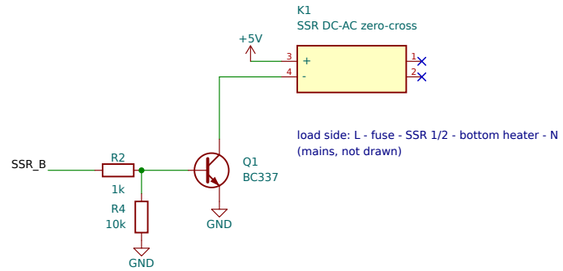
One or two controllers on the same A / B pair (own addresses).
No answer: swap A / B. No 120 ohm terminators below 8 devices.
RGND not connected. Auto-direction module: DE / RE open.

Thermocouples (local channels)



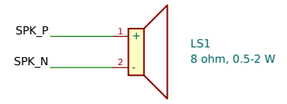
Shared SCK / SO, one chip select per converter.
MAX6675 modules fit the same way.

SSR drivers and SSRs (local channels)



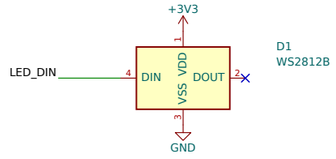
Q: BC337 or 2N2222 (TO-92 C-B-E). The 10k base pull-down keeps the SSR off during reset and flashing.
SSR + input: 5 V from P1-1 (VIN).

Alarm sound (speaker on P4)



IO26 drives the CYD's SC8002B amplifier, its bridged output is P4: SPK- is not ground, connect the speaker only between SPK+ and SPK-. A small 8 ohm speaker or a passive piezo. 2.7 kHz beeps (setting safety.sound; touch: Settings > System > Alarm sound).

Status LED (external WS2812 on IO17)



Powered from 3V3: the 3.3 V data level is then in spec.
Colours as on the S3 (green idle, blue running, ...).

Modules connected with cables; pins: src/board_cyd.h

Title: **TempCtl Duo - CYD C: RS485 + two local channels**

env cyd-2xlocal-2x485, firmware 1.0.39

Size: A3 | Date: 2026-10-13 | Rev: 1.0.39 | Sheet: 1/1

File: TempCtl-Duo-CYD-C.kicad_sch